

Effects of different non-driving-related-task display modes on drivers' eye-movement patterns during take-over in an automated vehicle



Xiaomeng Li ^{a,*}, Ronald Schroeter ^a, Andry Rakotonirainy ^a, Jonny Kuo ^b, Michael G. Lenné ^b

^a Centre for Accident Research and Road Safety-Queensland (CARRS-Q), Institute of Health and Biomedical Innovation (IHBI), Queensland University of Technology, Kelvin Grove, Queensland 4059, Australia

^b Seeing Machines Ltd., Fyshwick, ACT, Australia

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ABSTRACT

Highly automated vehicles relieve drivers from driving tasks, allowing them to engage in non-driving-related-tasks (NDRTs). However, drivers are required to take over control in certain circumstances due to the limitations of highly automated vehicles. This study focused on drivers' eye-movement patterns during take-overs when an NDRT (watching videos) was presented via a head-up-display (HUD) and a mobile device display (MDD), compared to no NDRT as the baseline. The experiment was conducted in a high-fidelity driving simulator with real-world driving videos scenarios. Forty-six participants took part in the experiment by completing three drives in three counterbalanced conditions (HUD, MDD and baseline). A take-over-request was issued towards the end of automated driving requesting drivers to stop the NDRT and take over control. Eye-movement data including pupil diameter, blinks, glance duration and number of AOI (Area of Interest) were collected and analysed. The results show that during automated driving, drivers were more engaged in the MDD NDRT with smaller pupil diameter and shorter glance duration on the front scenario compared to the HUD and baseline modes. The number of AOI was reduced during automated driving in both MDD and HUD modes. The take-over-request redirected drivers' visual attention back to the driving task from NDRT by increasing drivers' pupil diameter, glance duration and number of AOI. However, the effect of MDD NDRT on pupil diameter and glance duration continued even after the take-over-request when the NDRT was terminated. The study demonstrated HUD is a better display to help maintain drivers' attention on the road.

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1. Introduction

In recent years, automated vehicles have been developed at a fast pace with promising potential to improve road safety, mobility, and provide environmental benefits (Fagnant & Kockelman, 2015). The SAE International (Society of Automotive Engineers) provided a taxonomy (SAE 2018) classifying vehicle automation into six incremental levels (Level 0–5). L0 refers to the fully manual vehicles, which accounts for the majority of vehicles on the road today, while some luxury cars provide

* Corresponding author.

E-mail address: xiaomeng.li@qut.edu.au (X. Li).

L2 support features (e.g. adaptive cruise control systems combined with lane-keeping assistance), and L3 and above being either in development or testing stage, or operating in highly controlled environments.

There is a critical paradigm shift between SAE L2 and L3, where systems are no longer regarded as driver support features, but automated driving features. It is widely accepted that the introduction of L3 automated vehicles fundamentally changes the traditional concept of a vehicle driver which introduces unresolved human factors challenges.

The human factor challenge for L3 and L4 lies in that humans might be either *required* to take over control when the automated system requests it (L3), e.g. during system failures, or *desired* to take over control (L4), e.g. when the vehicle reaches its limitations, or Operational Design Domain (ODD). The difference between L3 and L4 lies in the L4 automation's ability to conduct a manoeuvre that brings the vehicle to a safe state if the human driver does not take over control, e.g. stop on the side of the road.

In a L3 + automated vehicle, drivers no longer need to monitor the driving task (the SAE refers to this as Object and Event Detection, or OED) and thus their role shifts from a driver to a passenger, enabling them to engage in other tasks, so called non-driving-related tasks (NDRTs). The NDRTs are not defined, but conceivably include watching a video, reading the news or writing an email, etc. Up until now, a large body of research has investigated the various types of NDRTs that drivers may engage in and their impacts on drivers' take-over performance (Merat, Jamson, Lai, & Carsten, 2012; Zeeb, Härtel, Buchner, & Schrauf, 2017; Naujoks, Höfling, Purucker, & Zeeb, 2018).

1.1. Non-driving-related tasks in automated driving

One of the main goals of vehicle automation systems is to increase drivers' comfort by reducing the demand imposed by the driving tasks. A tremendous change that automated vehicles bring to drivers is that they can now primarily focus on NDRTs during automated driving. Studies have shown that as the level of automation increases, drivers are more likely to engage in NDRTs (De Winter et al., 2014). Drivers are attracted by NDRTs even when they are informed that they remain responsible for monitoring the automated driving system (Dogan et al., 2017). Therefore, a large number of studies have examined drivers' performance in an automated vehicle while providing them with different kinds of NDRTs.

Some of the studies applied the standardised NDRTs, such as the delayed digit recall (n-back task) or Surrogate Reference Task (SuRT), to test the effect of NDRTs on drivers' performance in automated driving (Radlmayr, Gold, Lorenz, Farid, & Bengler, 2014). These tasks are designed to attract drivers' eyes- and/or mind-off road, and they have been identified by the ISO (the International Organization for Standardization) as standardized methods to assess driver attentional demand due to the use of in-vehicle information and communication devices¹. The standardised NDRTs have been proved to be reliable to create comparable cognitive and/or visual-manual distractions (Radlmayr et al., 2014; Radlmayr, Fischer, & Bengler, 2018).

However, the NDRTs should not be designed specifically to distract drivers, but to be representatives of real-life tasks that drivers are inclined to engage voluntarily under real-world conditions (Naujoks, Befelein, Wiedemann, & Neukum, 2017). Clark and Feng (2017) observed drivers' voluntary engagement in NDRTs whilst various non-driving-related activities were provided, such as reading, grooming, electronic device use, conversing with another, conversing with self, horseplay, and listening to music. Other research used common daily activities as the NDRTs when studying take-over performance. For instance, Yoon and Ji (2019) had participants interacting with an entertainment console, watching a video and interacting with a smartphone, and Dogan, Honnêt, Masfrand, and Guillaume (2019) used writing emails and watching videos as NDRTs when assessing drivers' obstacle avoidance performance after take-over.

In our study, we choose watching videos as the NDRT as it is one of the most common activities that drivers may voluntarily engage in to pass time and it has been widely considered in previous studies (Blommer et al., 2015; Dogan et al., 2019; Yoon and Ji, 2019).

1.2. Effects of NDRTs on take-over performances

A take-over can typically be described through the following steps: 1. *Automated driving*, during which the driver may undertake an NDRT, 2. the AV's *Take-Over Request (TOR)*, followed by 3. a *transition phase*, during which the driver states transitions towards manual control, and the AV is in take over mode, and lastly 4. *manual driving*, when the driver finally controls the vehicle (Marberger et al., 2017; SAE 2018). The NDRTs' effects on such a take-over largely depend on the different types and modalities of the NDRT, the complexity and criticality of the take-over situation, the automation level, and lastly, how a positive or negative performance is measured.

Engaging in NDRTs was found to deteriorate drivers' take-over performance or take-over quality in terms of reaction time, speed and steering control stability, etc. Zeeb et al. (2017) examined the impact of both manual and cognitive NDRTs on drivers' ability to take over vehicle control. Both tasks were found to prolong drivers' reaction time and affect the steering manoeuvre. Merat et al. (2012) found drivers' speed adjustment ability after take-over was impaired due to the engagement in NDRTs prior to the TOR.

There are other studies, however, that did not find a significant role of NDRTs impacting on the take-over negatively. Several recent studies reported that the NDRTs did not affect drivers' take-over time (Zeeb, Buchner, & Schrauf, 2016; Dogan

¹ <http://agelab.mit.edu/delayed-digit-recall-n-back-task>; <https://www.iso.org/obp/ui/#iso:std:iso:ts:14198:ed-1:v1:en>

et al., 2019; Yoon and Ji, 2019). Moreover, some studies reported positive effects of engagement in NDRTs. For example, Naujoks et al. (2018) found that visual and mental demand associated with NDRTs decreased reaction time, suggesting that NDRTs helped drivers maintain alertness during partially automated drive. Neubauer, Matthews, and Saxby (2012) found that phone use during automation was associated with a faster braking response in the transition from automated driving to manual driving. Jarosch, Kuhnt, Paradies, and Bengler (2017) and Schömig, Hargutt, Neukum, Petermann-Stock, and Othersen (2015) both found that drivers stayed on a significantly lower fatigue/drowsiness level when they were engaged in NDRTs, which suggests that NDRTs could help drivers maintain alertness during automated driving.

Drivers' performances during take-over while engaging in NDRTs have been evaluated from various perspectives. In automated driving, driver's actual vehicle control behaviour is no longer available since the automation system controls the car. Therefore, eye-movement measures, which are made available in real-time by advanced eye-tracking techniques, have been largely applied in previous studies as the performance measures, e.g. as an index of drivers' mental status and visual attention. Especially when drivers are engaging in NDRTs, their visual attention distribution between the NDRTs and road situation reflects their readiness to take over control to some extent, and thus plays an important role in the take-over quality (Louw et al., 2019; Yoon and Ji, 2019).

The time for visual attention to redirect to the roadway-related area after a TOR is widely used as a measurement of drivers' reaction ability. Studies consistently reported that engagement in a visual-demanding NDRTs significantly extended drivers' "eyes-on-road" (or visual reaction) time (Zeeb et al., 2017; Yoon and Ji, 2019). Except for the visual reaction time, the eye-movement pattern and visual attention allocation among different areas of interest (AOIs) were also explored while drivers engaged in NDRTs in automated driving (Zeeb, Buchner, & Schrauf, 2015; Louw & Merat, 2017). Zeeb et al. (2015) categorized drivers into different risk groups according to their gaze behaviours, and further predicted the readiness to take-over the vehicle when a visual NDRT was provided. Merat et al. (2012) reported an increased blink frequency when drivers performed a NDRT (Twenty Questions Task) during automated driving, and this was accounted for the high workload demand induced by the NDRT.

Overall, the majority of studies tended to report results of a specific aspect of the eye-movement measurements (e.g. glance-related or blink-related metrics) regarding an overall driving situation (e.g. with vs without NDRTs in manual vs. automated driving). In contrast, there has been a paucity of research targeting the take-over process with a comprehensive analysis of multiple eye-movement measurements (e.g. pupil diameter, blink, glance, AOI, etc.).

1.3. Mobile device display and Head-up display

Using a mobile device such as mobile phone to present the NDRT has been commonly applied in prior studies (Naujoks et al., 2017). Mobile phones have become an indispensable part of people's life and drivers on road were found to frequently use their phones during driving (Oviedo-Trespalacios et al., 2019). As the vehicle automation increases, drivers will likely engage more in NDRTs with their phones. A survey by Pfleging, Rang, and Broy (2016) showed that several of the most wanted activities for automated driving involve a mobile device. Neubauer et al. (2012) investigated drivers' mobile phone use behaviour and driving performance in both simulated manual driving and automated driving conditions. Their study suggests that drivers in automated driving responded more frequently to the text messages compared to drivers in the non-automated driving. Using mobile devices during driving could increase both manual and cognitive workload and distract drivers' attention away from the forward road (Li, Yan, Wu, Radwan, & Zhang, 2016 & 2018). Although automated vehicles promise drivers more freedom in NDRTs, it is necessary to develop innovative approaches that allow for a safe engagement in the NDRTs.

Head-up display (HUD) is a technology that enables visual information to be displayed in the drivers' forward field of view as a transparent projection on the windscreen (see Fig. 1). Its idea is to reduce the degree to which drivers need to shift their gaze or move their head between tasks, e.g. when switching from scanning the outside road environment to obtaining information from typical in-vehicle displays in the centre console or the instrument cluster. By limiting gaze and head movements, HUDs can reduce drivers' eye-off-road time, and thus aid maintaining their visual attention on the forward roadway. As such, in-vehicle HUD applications have long been deemed with great potential to improve driving performance and safety.

In the context of manual driving, a number of studies have focused on the driving-related information that HUDs can be used to display. The studies have demonstrated that HUDs offer benefits over traditional panel-mounted displays in terms of driver response time (Liu, 2003; Liu & Wen, 2004), situation awareness (Liu, 2003), subjective workload (Liu, 2003) and user experience, etc. Specifically when drivers were performing a secondary task while driving, the HUD was reported to lead to reduced detrimental effects on driving performance (Wittmann et al., 2006). However, due to the safety critical nature of driving, only few studies have combined advanced HUD functions with NDRTs to create new driver experiences. Charissis, Naef, Papanastasiou, and Patera (2007) designed a HUD interface that can show phone-call-related information, SMS and email message to aid driver awareness during in-vehicle mobile communications. Schroeter, Oxtoby, and Johnson (2014) applied gamification techniques with HUD to reduce drivers' boredom and risky driving. The safety concern in the trade-off between NDRTs and driving tasks constitutes a limitation for the application of HUDs in NDRTs in manual driving context.

In the context of emerging automated vehicles, HUDs represent a promising safer device for drivers to engage in NDRTs during automated driving. Both, Level 3 and Level 4, may benefit from a human driver taking control who maintains some

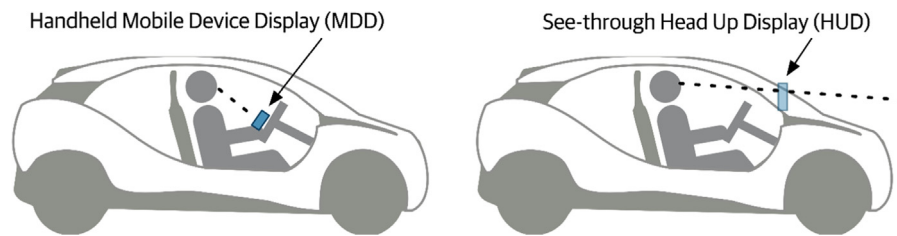


Fig. 1. Illustration of MDD and HUD.

level of intermittent attention on the road situation. If utilized as a primary display in automated vehicles, HUDs may have the advantage to better facilitate such intermittent attention.

Studies that tested HUD as an information display mode in the field of automated driving are scarce. [Utsumi, Kawanishi, Nagasawa, Satou, Uchikata, and Hagita \(2018\)](#) used HUD to display steering direction information to guide drivers when an emergency situation appeared in automated driving. [Schömig et al. \(2018\)](#) designed an Augmented Reality Head-up Display (AR-HUD) to support drivers during conditionally automated driving by displaying vehicle speed, speed limit, reason for TORs or road-based regulations. Moreover, there is a lack of studies that combine HUD with NDRTs in automated driving when engagement in NDRTs is allowed.

1.4. Objectives and research questions

The effect that different display modes of NDRT (HUD vs mobile device display, MDD) have on drivers' visual performance before and after a TOR, remains largely unknown. The investigation of the present study aims to address this research gap. Specifically, this study aims to investigate the following three questions: (1) What are the effects of different NDRT modes (HUD, MDD and baseline) on drivers' eye-movement patterns in automated driving? (2) What are the changes in drivers' eye-movement performance when a TOR is issued? (3) How does the NDRT before the TOR affect drivers' eye-movement performance after the TOR?

Investigation into this study will help automated vehicle designers to develop HUD functions that consider the drivers' need or desire to engage in NDRTs during automated driving. The study further contributes to a better understanding of how drivers' visual performance is affected by different NDRT display modes.

2. Methodology

2.1. Experiment apparatus and driving scenario

The CARRS-Q Advanced Driving Simulator was used in this study. This simulator consists of a real vehicle cabin (Holden Commodore) with automatic transmission, a surround-sound system for engine and environmental noise, and a six degree of freedom motion platform that can move and twist in three dimensions (see [Fig. 2](#)). The vehicle is surrounded by three front-view projectors providing a 180 degrees high-resolution field view to drivers. Wing mirrors and the rear view mirror are replaced by LCD monitors to simulate the side and rear-view images.

The Driver Monitoring System (DMS) developed by Seeing Machines Limited was used to collect drivers' eye-movement data during the experiment. It uses a sensor bar (with an infrared camera embedded in the centre and a pair of pulsed infrared lights on both sides) installed on the front dashboard of the vehicle (above the steering wheel and instrument cluster for a least obstructed view) to monitor drivers' head and eye-movement as well as facial expressions/gestures, as shown in [Fig. 3](#). The recorded videos were further processed by Seeing Machines' algorithms to output the detailed position, movement, orientation information of head/face/eye.

To create a visually more realistic and stimulating automated driving environment compared to the typical simulated 3D graphics of the simulator software, the experiment scenarios used a custom tool developed in Unity 3D, called Immersive Video-based Automated Driving (IVAD) Simulator ([Gerber, Schroeter, & Vehns, 2019](#)). IVAD uses seven videos recorded during real-world driving that are played back in an immersive, visually surrounding way (see [Fig. 4](#)): three at 60 degrees out the front to create a forward 180 degree view when projected on the three front projection screens ([Fig. 4.e-g](#)), three to represent three different rear mirror views played back in the simulators LCD screens ([Fig. 4.a-c](#)), and one screen-capture video of the navigation played back in the in-vehicle navigation HMI (Human-Machine-Interface, see [Fig. 4.d](#)). Driving sounds were recorded through the in-vehicle cameras and also played back in the simulation. IVAD underwent pilot and user studies to evaluate and improve the tool prior to conducting this study ([Gerber et al. 2019](#)).

During the recording of the trips used in this study, the vehicle was controlled manually to drive in a suburban area of Brisbane, Australia. The driving route, time of day, traffic density, driving style, etc. was the same for each trip. To mimic the performance of an automated vehicle, we followed the speed limits, drove passively in terms of following distances, and highly anticipatory in terms of lane selections.



Fig. 2. CARRS-Q driving simulator.

2.2. Experiment design

The study used a within-subject measures design. The within-subject factor is the different display modes of entertainment videos, which includes a mobile device display (MDD), a head-up-display (HUD), and no video display as the baseline condition. Drivers could self-select two episodes of a TV show to be watched respectively in the two display modes during the automated driving. Self-selection was chosen so that participants would voluntarily engage in the video content, as they would in the real-world. In the MDD mode, a smartphone (iPhone 6) was provided to drivers to watch the TV show and drivers could hold the smartphone in any position they felt comfortable. In HUD mode, the TV show was displayed on the front projection screen with 76 cm × 43 cm size and 61.5% transparency at a distance of approximate 3.5 m from the driver's eye (see Fig. 5).

All participants experienced three different trips of the same driving route, during which they experienced each of the three conditions. The order of three trips was counterbalanced among all participants to reduce the order effect and learning effect. Each trip was approximately 15 min long and represented a typical commuter trip to an average distanced suburb in Brisbane. To enrich the automated driving experience of participants, the trips included driving on both inner-city roads and highways. A controlled TOR was released at the end of automated driving (at around 13 min) when the vehicle exited the highway and entered a suburban road. The TOR design represents a realistic situation of an AV exiting its operational design domain and requiring human drivers to take-over (SAE 2018).

The controlled TOR was issued through audible (a series of “beep” sound) and visual warnings (numbers counting down from 10 s) on the navigation HMI. In HUD mode, the TV show stopped and disappeared automatically when the TOR was released. In the MDD mode, participants were required to stop the video entertainment task manually when they received

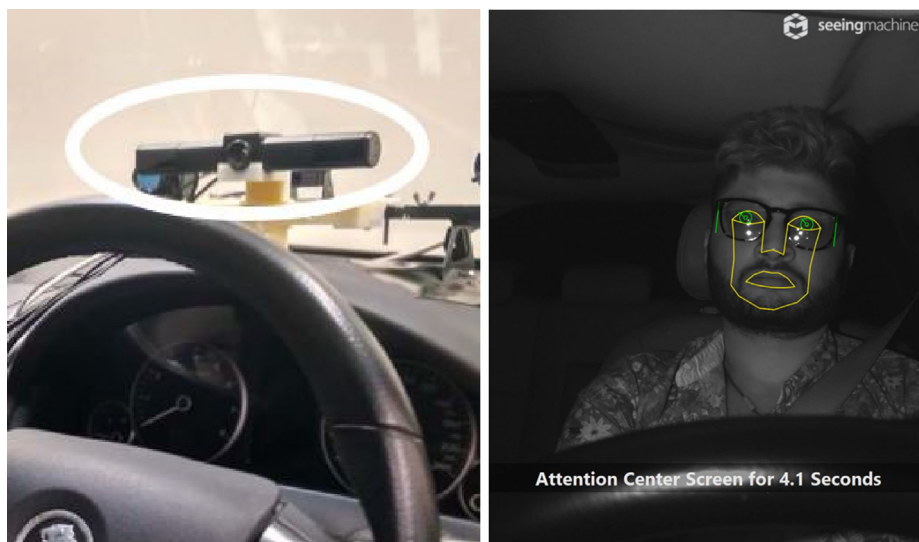


Fig. 3. Seeing Machine's DMS dashboard mounting (left) and resulting driver view (right).



Fig. 4. Scenario recorded from real-world driving.



Fig. 5. Scenario with HUD.

the TOR to prepare for the take-over. The volume of TOR “beep” sound was controlled higher than the entertainment video sound to ensure drivers received the TOR clearly. Once the TOR was issued, participants were required to take over control of the vehicle within 10 s and keep driving until the end of the scenario.

After the participants took over control of the vehicle, participants were asked to “shadow drive” using the steering wheel, accelerator and brake paddle based on the vehicle trajectory depicted in the driving video. This was introduced since participants can not actually control the vehicle in the video, a limitation of the IVAD Simulator. As the study only focuses on drivers’ eye-movement performance, this limitation was outweighed by the visual fidelity and scenario richness that the IVAD provided. Additionally, participants were given extended familiarisation of “shadow driving” prior to the research drives to allow overcoming the initial awkwardness identified in pilot studies (Gerber et al. 2019).

2.3. Participants and experiment procedure

Forty-six participants including 23 females and 23 males took part in the experiment. The participants aged between 20 and 62 years old, with an average age of 32 years old and a standard deviation of 8.4 years. All participants held a valid driver’s licence in Australia. They were recruited through social media posts, university web page advertisements, and snowball sampling.

Before the experiment, all participants signed a consent form to participate and were informed that their participation was entirely voluntary and the data collected would remain anonymous and be reported in an aggregate manner only. They were introduced to the experiment procedure (e.g. that they could quit the experiment at any time in case of motion sickness or any discomfort) and what they would experience during the experiment, including the capabilities of the automated vehi-

cle (L3), the NDRT, the take-over request and the “shadow” driving. Before stepping into the simulator, each participant also chose two episodes from their favourite TV show to watch during two of their research drives.

Following the preparation, all participants completed an approximately five minutes familiarization drive in the simulator to experience the process of automated driving, take-over and “shadow” driving. They were advised to do the “shadow” driving until comfortable with the concept, followed by a motion sickness questionnaire to assess their suitability to continue the study and subsequent three research drives (a requirement of our simulator's standard operating procedure).

Overall, it took about 1 h to complete the three drives, including interruptions to reset and a brief break in between. After completing the experiment, each participant received an AUD 70 gift voucher as compensation for their participation. The study was conducted in accordance with the Australian Code for the Responsible Conduct of Research (QUT Ethic approval number 1700000425).

2.4. Data collection and analysis

The eye-movement data were recorded at a sampling rate of 60 Hz by Seeing Machines' Driver Monitoring System (DMS). This paper focused on drivers' eye-movement performance during the take-over process. When the TOR was released, participants were required to take over control within 10 s. Therefore, eye-movement variables were extracted 10 s before the releasing of TOR and 10 s after that respectively. The variables included (1) pupil diameter (the average diameter of left and right pupils, in mm), (2) blink rate (the number of blinks per second), (3) gaze duration on front scenario (the accumulated time of gazes that located on the front screen, in s), and (4) number of AOI. The AOI refers to the specific area that drivers' gaze located. In this study, we identified 9 areas of interest, namely front screen, right screen, left screen, centre console, instrument panel, rear mirror, right mirror, left mirror and navigation HMI. Prior to the experiment, these areas have been modelled in a 3D environment relative to the DMS camera.

As the variables followed the assumptions of parametric tests, paired *t*-test was used to compare the difference of eye-movement variables before and after TOR. Considering the repeated measures design of the study, Generalised Linear Mixed Models (GLMMs) were applied to examine the effect of video display modes on drivers' eye-movement variables before and after TOR, respectively. As an extension of linear models, GLMMs allow for modelling dependent variables from different distributions and account for both fixed and random effects. If the effect of the video display modes was at a significant level ($p < 0.05$), pairwise contrast was further conducted for each two conditions.

3. Results

3.1. Pupil diameter

The descriptive and statistical test results of pupil diameter within 10 s respectively before and after the TOR in three video display conditions are listed in Table 1. The GLMMs results show that before the TOR, different video display conditions had a significant effect on drivers' pupil diameter ($F(2, 37) = 26.761$, $p < 0.001$). As shown in Fig. 6.a, the pupil diameter was largest in HUD condition and smallest in MDD condition. The pairwise comparisons showed that the pupil diameter in MDD condition was significantly smaller than that in baseline condition ($p < 0.001$) and HUD condition ($p < 0.001$).

Even though the video display stopped when the TOR was issued, the model results indicate that it still significantly affected drivers' pupil diameter after the TOR ($F(2, 37) = 27.946$, $p < 0.001$). More specifically, the pupil diameter was smaller in MDD condition compared with baseline condition ($p < 0.001$) and HUD condition ($p < 0.001$), and the pupil diameter in baseline condition ($p < 0.05$) was also smaller than that in HUD condition (see Fig. 6.b).

To examine the changes before and after the TOR, *t*-tests were conducted and the results show that in all three video display conditions, drivers' pupil diameter dilated significantly after the TOR (Baseline: $t = -6.979$, $p < 0.001$; MDD: $t = -8.167$, $p < 0.001$; HUD: $t = -5.980$, $p < 0.001$) (see Fig. 6.c).

3.2. Blink rate

Table 2 shows the descriptive and statistical tests results of blink rate within 10 s respectively before and after the TOR in the three conditions. The GLMMs results show that there was no significant difference among the three video display conditions both before and after the TOR. Additionally, paired *t*-tests show that there was no significant change of blink rate before and after the TOR in all three conditions.

3.3. Glance duration

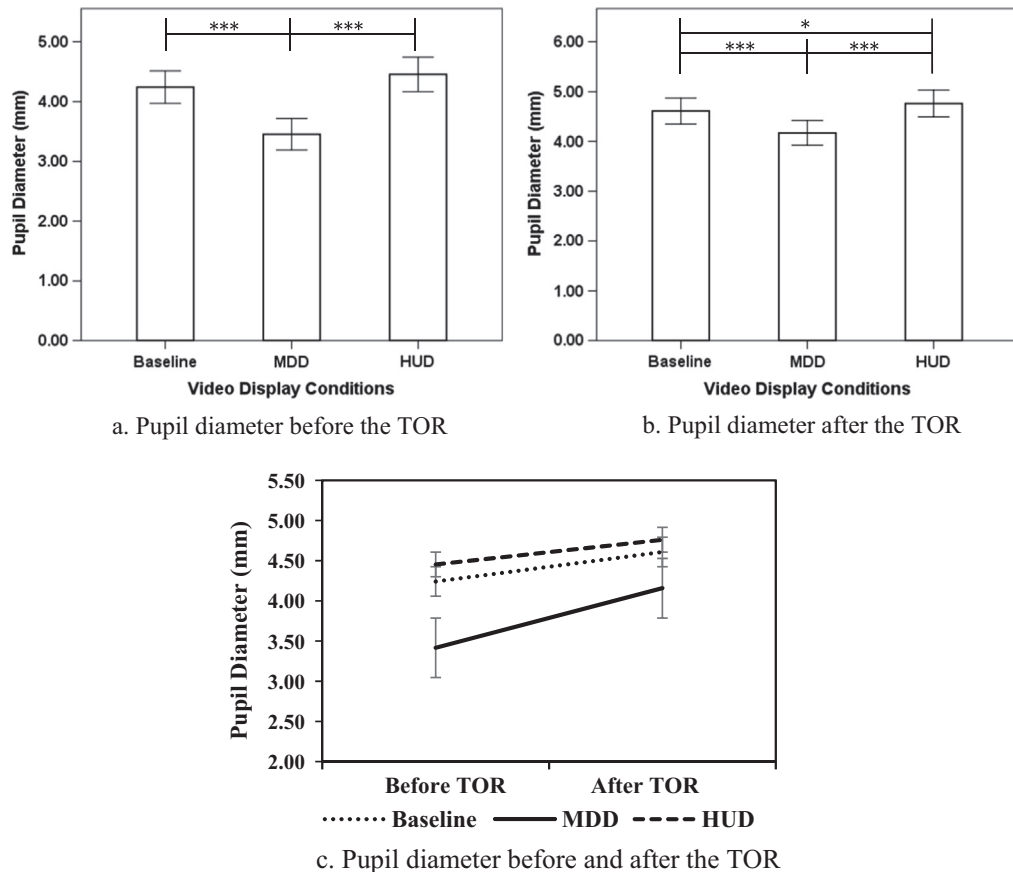
The descriptive and statistical tests results of glance duration on front scenario before and after the TOR in three video display conditions are listed in Table 3. The video display conditions had significant effects on drivers' glance duration on front scenario both before ($F(2, 46) = 77.946$, $p < 0.001$) and after the TOR ($F(2, 46) = 41.818$, $p < 0.001$). Before the TOR, drivers had longer glance duration on front scenario in baseline condition ($p < 0.001$) and HUD condition ($p < 0.001$) than in MDD condition, see Fig. 7.a. The tendency continued after the TOR, i.e., the glance duration on front scenario was still longer

Table 1

Descriptive and statistical tests results of pupil diameter.

Conditions	Before TOR		After TOR		df	t-ratio	95% CI
	Mean	SD	Mean	SD			
Baseline	4.27	0.94	4.62	0.90	45	−6.979***	(−0.45, −0.25)
MDD	3.42	0.71	4.14	0.81	36	−8.167***	(−0.90, −0.54)
HUD	4.44	0.95	4.76	0.90	47	−5.980***	(−0.42, −0.21)
df	2		2				
F-ratio	26.761***		27.946***				

*** represents a significance level of 0.001

**Fig. 6.** Pupil diameter in different conditions.**Table 2**

Descriptive and statistical tests results of blink rate.

Conditions	Before TOR		After TOR		df	t-ratio	95% CI
	Mean	SD	Mean	SD			
Baseline	0.56	0.42	0.45	0.24	39	1.596	(−0.02, 0.19)
MDD	0.60	0.56	0.47	0.31	31	1.824	(−0.02, 0.28)
HUD	0.52	0.46	0.44	0.31	40	1.266	(−0.04, 0.17)
df	2		2				
F-ratio	0.854		0.252				

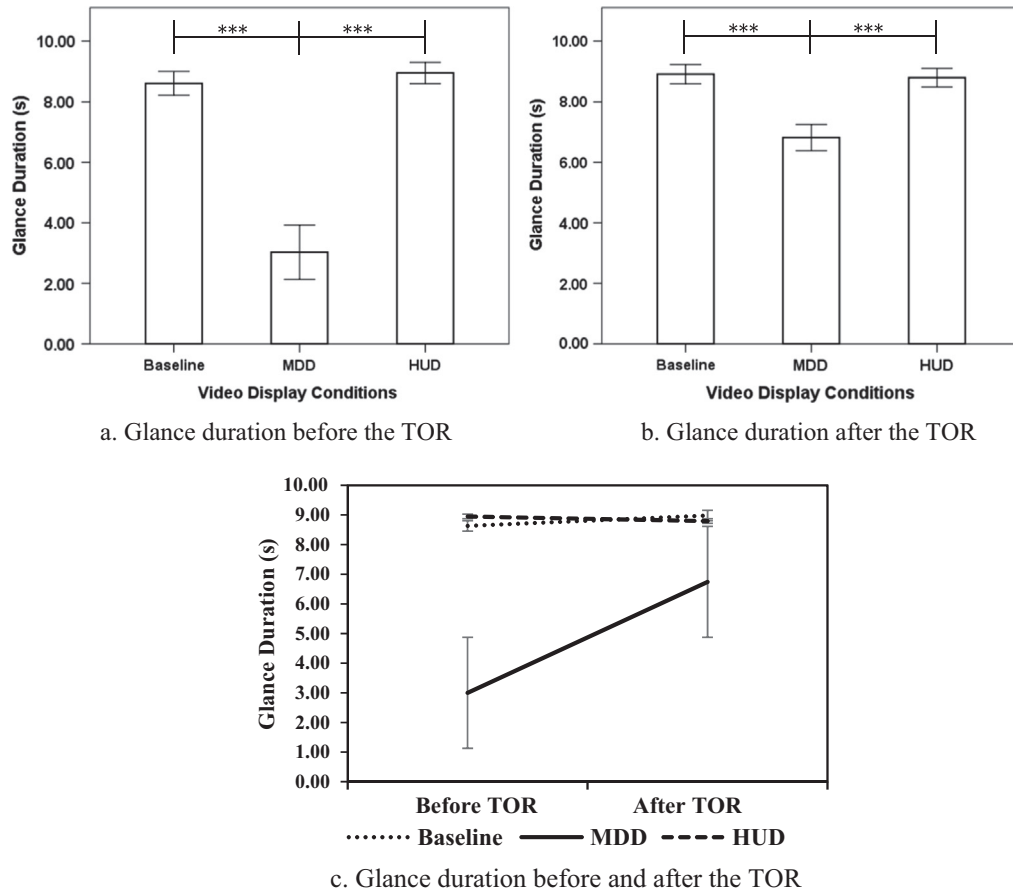
in baseline ($p < 0.001$) and HUD ($p < 0.001$) conditions compared with MDD condition, see Fig. 7.b. Additionally, significant increase in glance duration on front scenario was observed in MDD condition after the TOR was issued ($t = -8.415$, $p < 0.001$), see Fig. 7.c.

Table 3

Descriptive and statistical tests results of glance duration on front scenario.

Conditions	Before TOR		After TOR		df	t-ratio	95% CI
	Mean	SD	Mean	SD			
Baseline	8.63	1.23	8.98	0.86	45	−2.006	(−0.70, 0.00)
MDD	3.00	0.44	6.74	0.22	45	−8.415***	(−4.64, −2.85)
HUD	8.95	1.18	8.80	1.03	45	0.755	(−0.26, 0.57)
df	2		2				
F-ratio	77.946***		41.818***				

*** represents a significance level of 0.001

**Fig. 7.** Glance duration on front scenario in different conditions.

3.4. Number of AOI

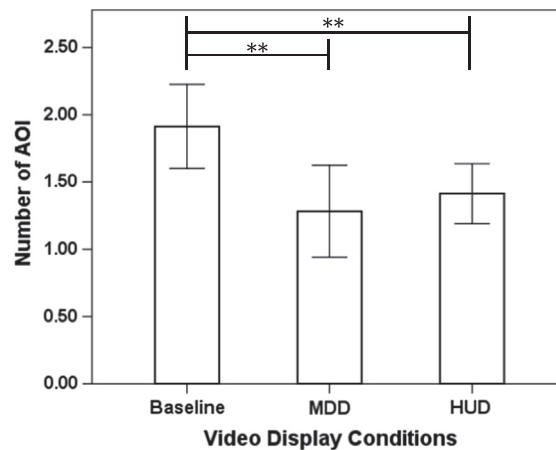
Table 4 shows the descriptive and statistical tests results of AOI number within 10 s before and after the TOR in three video display conditions. The numbers of AOI were significantly different among different video display conditions before the TOR was issued ($F(2, 46) = 4.866$, $p < 0.01$). Drivers watching videos through MDD ($p < 0.01$) and HUD ($p < 0.01$) had significantly fewer number of AOI compared with baseline driving (see Fig. 8.a). However, the difference between conditions was not significant after the TOR. When the TOR was issued, the number of AOI increased significantly in two conditions with video display, i.e. MDD ($t = -3.486$, $p < 0.01$) and HUD ($t = -2.506$, $p < 0.05$), while no significant change was observed in baseline condition (see Fig. 8.b).

Table 4

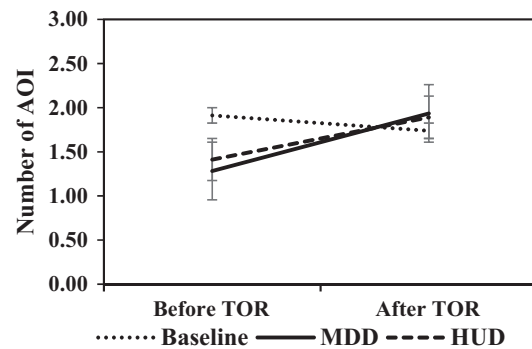
Descriptive and statistical tests results of number of AOI.

Conditions	Before TOR		After TOR		df	t-ratio	95% CI
	Mean	SD	Mean	SD			
Baseline	1.91	0.16	1.74	0.16	45	0.984	(-0.182, 0.530)
MDD	1.28	1.15	1.93	1.10	45	-3.486**	(-1.03, -0.28)
HUD	1.40	0.74	1.91	1.25	46	-2.506*	(-0.921, -0.101)
df	2		2				
F-ratio	4.866**		0.454				

** represents a significance level of 0.01; * represents a significance level of 0.05



a. Number of AOI before the TOR



b. Number of AOI before and after the TOR

Fig. 8. AOI number in different conditions.

4. Discussion

In this study, we investigated drivers' eye-movement patterns during the take-over process when different modes of video display (MDD, HUD, and no video as baseline) were provided in automated driving. Analysis of the pupil diameter, blink rate, glance duration (on front scenario) and number of AOI indicates that video entertainment as a non-driving-related-task (NDRT), had significant effect on drivers' eye-movement performance both before and after the TOR was issued. The take-over-request also played an important role in adjusting drivers' visual attention and visual search strategy between the NDRT and driving task.

Regarding the two video display modes, eye-movement results show that drivers focused more on the NDRT presented through MDD compared with HUD. The pupil diameter was significantly larger in HUD condition than MDD condition. Previous studies have reported consistently that pupils dilate in response to an increased task demand (Tsai, Viirre, Strychacz, Chase, & Jung, 2007; Čegovnik, Stojmenova, Jakus, & Sodnik, 2018; Li, Rakotonirainy, Yan, & Zhang, 2018). Once the auto-

mated vehicle relieves drivers from the driving task, it is quite likely for a mobile device to draw drivers' attention with NDRTs given the level of trust drivers have on the automation system (De Winter et al., 2014; Banks, Eriksson, & O'donoghue, J., Stanton, N.A., 2018). In contrast, when the NDRT was presented through HUD and overlapped with the forward scenario, drivers are more likely to assign their attention between the NDRT and the road environment. In other words, drivers were probably performing self-initiated multi-tasks in HUD driving condition, and thus the task demand was higher than that in MDD condition.

Pupil diameter has been identified as a sensitive variable to changes in mental workload induced by a variety of secondary tasks, such as auditory task (Tsai et al., 2007), visual task (Annerer-Walcher, Körner, & Benedek, 2018), or auditory-verbal-vocal task (n-back) (Niezgoda, Tarnowski, Kruszewski, & Kamiński, 2015). Compared with the primary driving task, drivers' pupil diameter increased when they performed a secondary distraction task simultaneously (Tsai et al., 2007; Niezgoda et al., 2015; Annerer-Walcher et al., 2018). Differently in this study, the role of primary task and secondary task exchanged in automated driving: the changing road environment "distracted" drivers' attention from watching the entertainment video on front screen and increased their mental workload in HUD driving condition. Given that no blink rate difference was observed among different driving conditions, the result also supported previous finding that pupil diameter was more sensitive than blink rate in measuring the changes in mental workload (Niezgoda et al., 2015). Nevertheless, it should be noted that pupil diameter is also sensitive to the environmental lighting conditions. In the scope of this study, the simulator experiment allows a stable lighting intensity within the simulation room. However, it is difficult to guarantee such stable conditions in a real-world driving, which may limit the application of pupil diameter as a reliable measurement.

The HUD mode was designed to provide drivers with entertainment task while maintaining a certain level of their visual attention on road. Results from glance duration and pupil diameter indicate that compared with the MDD, the HUD did attract drivers' visual attention on forward roadway, and the increased information input increased drivers' mental workload. However, the NDRT still limited the areas where drivers collected visual information, as the numbers of AOI were significantly smaller in MDD and HUD conditions compared with baseline condition. The result was consistent with the finding of (Zeeb et al., 2015) that when a visually demanding secondary task (e.g. texting, or internet search) was provided to drivers during automated driving, the glance duration revealed a strong focus on the secondary task. Another explanation could be approached through prior research that has suggested there was a threshold of glance duration at non-attention demanding targets in normal driving (Green, 1999). For example, drivers were hesitant to go for over 2 s without information from the roadway (Rockwell, 1972; Green, 1999). This helps explain our results that drivers were inclined to shift visual focus between several different AOIs (e.g., instrument panel, left/right scenario, left/right/rear mirror) in a monotonous automated driving, but they tended to focus their attention on fewer areas while watching the video on either display conditions.

Several changes in the eye-movements showed that the take-over request was effective in redirecting drivers' attention back to the loop, and the effect was especially remarkable when drivers were watching videos through MDD. Firstly, drivers' pupil diameter dilated significantly after the TOR in all three conditions, indicating an increase of mental workload when drivers were required to take over control of the vehicle. Secondly, a significant increase in the glance duration on front scenario was observed in MDD condition after the TOR, which means drivers' visual focus was shifted from the NDRT to the forward roadway. Thirdly, the number of AOI increased significantly in both MDD and HUD conditions after the TOR. These changes can be explained in that the 10 s following a controlled TOR is a phase that involved perception, reaction and action. Procedures that drivers may perform within this stage included perceiving the ambient road and traffic situation, collecting necessary visual information, making the decision of when and how to take over, and taking actions (hands on the steering wheel and/or foot on brake/accelerator paddle). Even though the NDRT limited drivers' visual interest area to a great extent, when a TOR signal was received, the drivers' prior driving experience prompted them to collect vital road/traffic information from a variety of AOIs quickly while preparing to take over control of the vehicle.

A notable finding from the study is that the effects of MDD on drivers' eye-movement continued during the take-over process even after the NDRT was suspended. It was within expectation that the TOR could bring driver' attention back to the driving task, and accordingly, significant increases in pupil diameter, glance duration on front scenario and number of AOI were observed in all three drives. However, the pupil diameter and glance duration on front scenario after TOR were still smaller in MDD condition compared with the other two conditions. Previous studies reported that drivers with well-adapted allocation of visual attention between driving and secondary tasks tended to make regular glances at the roadway, and gained enhanced situation awareness compared to drivers focusing mostly on the secondary task (Zeeb et al., 2015). In the present study, drivers in the MDD condition engaged more in the NDRT during automated driving and had lower visual attention on the front roadway compared with the other conditions. Therefore, when the TOR was issued, drivers in the MDD condition had comparatively the least situational information. Consequently, they need to allocate attention among several information sources (i.e. increasing the number of AOI) to adequately perceive the road environment and update the information.

5. Limitations

Some methodological limitations of the present study should be acknowledged.

Firstly, the study used real-world driving videos as the experiment scenario instead of the scenario developed with the driving simulator. This method has been applied in many studies given that real-world videos can significantly enhance

the participants' immersion and sense of reality (Lu, Coster, & De Winter, 2017, Chen, Zhuang, Cui, & Ma, 2019, Wang, Wang, Li, & You, 2019). However, it introduces technical limitation as the simulator could not response to drivers' operation after they take over control, and participants have to perform the "shadow" driving.

Secondly, no take-over performance measurements (i.e., take-over time, speed control and lateral position control stability) were included in the study due to the technical limitation that the driving simulator's kinematics sensors and actuators were not connected to the driving scenario. Though the present study focused on drivers' eye-movement analysis, in future studies additional video perspectives recording drivers' behaviours more widely beyond their face, and drivers' physiological data such as foot electromyography data could be included to obtain the take-over reaction time.

Thirdly, we did not quantify drivers' visual attention (e.g. glance duration) on the NDRT in this study. In HUD condition, as the HUD area overlapped with the front scenario with a certain transparency, when drivers watched the front screen, it is unlikely to differentiate the attention drivers paid on the entertainment video and on the road situation. In terms of the MDD condition, as we did not require drivers to use the mobile phone at a specific position, it is thus difficult to define an AOI of mobile phone to extract drivers' glances within the area. However, it is suspected that drivers' visual attention on the NDRT should be negatively correlated with their situation awareness on the road situation. In future study, drivers' situation awareness during the take-over process could be further examined when related measurements are available.

6. Conclusion

Using a high-fidelity driving simulator and scenarios recorded from real-world driving, the study created an automated driving environment for participants to investigate their eye-movement patterns during the take-over process from automated driving to manual driving. A common daily activity (watching videos) was used as an NDRT for drivers to engage in during automated driving. The NDRT was designed to present through different modes: HUD, mobile display device (MDD), and a baseline condition with no NDRT. The study put forward the following three research questions, for which the summary of findings and results are discussed in turn:

- (1) What are the effects of different NDRT modes (HUD, MDD and baseline) on drivers' eye-movement patterns in automated driving?

During automated driving, it was found that drivers tended to engage more in the video displayed through MDD, as the pupil diameter was smaller and the total glance duration on the front scenario was shorter than those in HUD and baseline conditions. Additionally, the NDRT in both HUD and MDD conditions significantly reduced the number of AOI compared with baseline condition during automated driving.

- (2) What are the changes in drivers' eye-movement performance when a TOR is issued?

The TOR increased drivers' pupil diameter (in MDD condition), glance duration on front scenario (in MDD condition) and number of AOI (in both MDD and HUD conditions) significantly. The result implies that the TOR was effective in redirecting drivers' visual attention to the driving task, especially when they were engaged in NDRT through MDD.

- (3) How does the NDRT before the TOR affect drivers' eye-movement performance after the TOR?

Among the different NDRT display modes, it was found that the effect of NDRT on drivers' eye-movement continued after the TOR was issued, especially in MDD condition. After TOR when the NDRT was suspended, the pupil diameter and the total glance duration on the front scenario were still smaller in MDD condition compared with HUD and baseline conditions. Besides, it was found that drivers who engaged in NDRT before TOR had to increase the number of AOI after TOR to collect adequate visual information.

In summary, the study indicates that for the same type of NDRT, different display modes could lead to significantly different eye-movement patterns. Findings of the study implies that HUD has advantages over mobile device in displaying NDRTs as it helps drivers maintain visual attention forward and facilitates drivers to collect visual information from various sources. Additionally, when drivers were engaged in NDRT, they cannot be expected to take over adequately and timely following the TOR, and an enough time budget should be allowed for them to complete a series of preparation tasks (e.g. NDRT termination, visual attention shift, visual information collection, etc.) before taking-over.

CRedit authorship contribution statement

Xiaomeng Li: Investigation, Validation, Methodology, Formal analysis, Writing - original draft. **Ronald Schroeter:** Conceptualization, Investigation, Validation, Methodology, Writing - original draft, Funding acquisition. **Andry Rakotonirainy:** Conceptualization, Supervision, Writing - review & editing, Funding acquisition. **Jonny Kuo:** Data curation, Writing - review & editing. **Michael G. Lenné:** Conceptualization, Resources.

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